

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE NAME: INTERNET PROGRAMMING**  
**COURSE CODE: CSA43**

### **COURSE OUTCOME COVERED**

**CO-3:** Implement dynamic web applications by utilizing DOM-based client-side scripting and employing Java Servlets for server-side request processing and response handling. (L3)

Assessment Tool : Technical Presentation  
Weightage: 20%

**QUESTIONS: Total Marks: 20**

#### **1. Servlet Architecture & Workflow**

##### **Question:**

Present the **architecture of Java Servlets** and explain the complete **request–response cycle** with a neat diagram.

##### **Focus Areas:**

- Client–Server interaction
- Role of Web Container
- Request and Response objects

**Focus Areas:** Client–Server interaction, Role of Web Container, Request and Response objects.

##### **Answer**

Java Servlet is a server-side Java component used to create dynamic web applications. A servlet receives a request from a client, processes the request on the server, and sends a response back to the client. The servlet runs inside a web container such as

Apache Tomcat.

## **Servlet Architecture**

The main components are the client/browser, web server or web container, servlet, request object, response object, and database or other backend resources. The web container manages servlet creation, loading, execution and destruction.

## **Request–Response Cycle**

1. The user enters a URL or submits a form in the browser.
2. The browser sends an HTTP request to the server.
3. Tomcat receives the request and identifies the required servlet using its URL mapping.
4. The web container creates or obtains the servlet object and calls the appropriate method such as `service()`, `doGet()`, or `doPost()`.
5. The servlet reads data from `HttpServletRequest` and performs the required processing.
6. The servlet places the result into `HttpServletResponse`.
7. Tomcat sends the HTTP response back to the browser.
8. The browser displays the returned HTML or other response.

## **Role of Web Container**

The web container manages servlet life cycle, URL mapping, request and response objects, multithreading, session management, security-related processing and communication with the servlet.

## **2. Servlet Life Cycle Analysis**

### **Question:**

Explain the **Servlet Life Cycle** with a sequence diagram and analyze the role of `init()`, `service()`, and `destroy()` methods.

### **Focus Areas:**

- Life cycle phases
- Execution flow
- Performance implications

**Focus Areas:** Life cycle phases, Execution flow, Performance implications.

## **Answer**

The Servlet Life Cycle describes the stages through which a servlet passes from creation to removal. The web container controls the complete life cycle.

Servlet Class Loaded – Servlet Object Created – init() – service() – doGet() / doPost() –  
Repeated Requests – destroy() – Servlet Removed

### 1. init()

The init() method is called once when the servlet is initialized. It is normally used for one-time initialization, such as reading configuration or preparing resources. Because it is normally executed only once, initialization work should be kept efficient.

### 2. service()

The service() method is called for incoming requests. It determines the type of HTTP request and dispatches it to methods such as doGet() or doPost(). This is the main request-processing stage and may execute many times during the servlet's lifetime.

### 3. destroy()

The destroy() method is called once before the servlet is removed by the container. It can be used to release resources such as files, connections, or other objects.

## Performance Implications

Expensive one-time setup should generally be performed during initialization rather than for every request. Request-specific work belongs in service()/doGet()/doPost(). Proper cleanup in destroy() helps avoid resource leaks.

## 3. GET vs POST Method Evaluation

### Question:

Compare **GET and POST methods** used in form submission. Present scenarios where each method is appropriate.

### Focus Areas:

- Data transmission
- Security aspects
- URL limitations

**Focus Areas:** Data transmission, Security aspects, URL limitations.

### Answer

**GET Example:** A product search form where the user searches for “laptop”. The search terms can be represented in the URL.

**POST Example:** A registration, login, or product-creation form where information is submitted to the server.

**Important Security Point:** POST does not automatically make data secure. HTTPS should be used to protect data during transmission. Sensitive values should not be placed in URLs unnecessarily because URLs can be logged, cached, or shared.

#### 4. Session Tracking Techniques

**Question:**

Present different **session tracking mechanisms** (HttpSession, Cookies, URL rewriting). Evaluate their advantages and limitations.

**Focus Areas:**

- Session persistence
- Security concerns
- Use-case comparison

**Focus Areas:** Session persistence, Security concerns, Use-case comparison.

**Answer**

HTTP is stateless, meaning that each request is treated independently. Session tracking techniques allow a web application to associate multiple requests with the same user or session.

**HttpSession:** The servlet container can maintain a session and associate a session identifier with the client. It is commonly used for login state and shopping-cart information.

**Cookies:** Small pieces of data stored by the browser. They can hold preferences or, commonly, a session identifier that points to server-side session data.

**URL Rewriting:** The session identifier is added to links or URLs when cookies cannot be used. It is useful as a fallback but is less desirable when exposing identifiers in URLs creates additional risk.

#### 5. Cookies vs Sessions – Critical Analysis

**Question:**

Analyze and compare **Cookies and Sessions** in web applications. Present real-time use cases.

**Focus Areas:**

- Storage location
- Security risks
- Performance considerations

**Focus Areas:** Storage location, Security risks, Performance considerations.

**Answer**

Cookies and sessions are both used to maintain state between HTTP requests, but they differ mainly in where the information is stored. Cookies are stored by the client/browser, while session data is normally maintained on the server.

**Real-Time Use Cases**

1. **Online shopping:** A session can maintain the user's cart and login state while moving between pages.
2. **Banking or secure portals:** A server-side session can maintain authenticated state after login, while HTTPS protects the communication channel.
3. **Web personalization:** Cookies can store non-sensitive preferences such as language or theme settings.
4. **Student portal:** A session can maintain the authenticated student's state while accessing different pages.

**Conclusion:** Cookies are useful for small client-side state and preferences, while sessions are useful for maintaining server-side user state. In many web applications, the two work together: the browser stores a session identifier in a cookie and the server stores the associated session data.

**Code of Conduct:**

I Ch. Bhargav Ram (192425334) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

**RUBRICS FOR CALCULATION:**

| Technical Presentation     |            |                                                                    |                                                                |                                                          |                                                                     |
|----------------------------|------------|--------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weight age | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                     | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30         | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25         | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak   | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20         | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation       | Minimal or incorrect use of tools and data; poor visuals            |

|                 |    |                                                              |                                                        |                                                            |                                                                  |
|-----------------|----|--------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------|
| Communication   | 15 | Highly engaging, confident, and audience focused delivery    | Clear, organized, and professional communication       | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides            |
| Professionalism | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness |